

Auditable Keeps: Offline-Verifiable Decision Provenance for Self-Improving Agents

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Abstract—Self-improving agents repeatedly decide whether an observed change should become the parent of later work. A conventional rollout may preserve neither the evidence used for that decision nor a machine-checkable link from the evidence to the submitted artifact. We present a decision-provenance layer that records candidate–parent comparisons, binds artifacts and measurements by digest, and verifies lineage, coverage, recomputed statistics, and protocol conformance offline. We study it in reduced-window, modified-program RSI-Exam rollouts. The evidence supports the integrity and audibility contribution and exposes incomplete native provenance in the observed cohorts. It does not support a claim that the tested gate improves sealed reward: across ten blocked comparisons, three favor the instrument and seven favor the helper. Seven results come from machine-generated endpoint tables, the first Opus block comes from digest-bound capsule reward fields, and the remaining two Opus blocks come from a committed pre-probe summary rounded to three decimals. We therefore treat efficacy as refuted under the tested conditions and a proposed confirmation-limit mechanism as a hypothesis, not a causal result.

Index Terms—agent provenance, audit trails, self-improving agents, offline verification, research artifacts

I. INTRODUCTION

Long-horizon self-improvement benchmarks ask an agent to modify a working method, evaluate candidates on visible data, and eventually submit one artifact for sealed evaluation. RSI-Exam is one such benchmark: a task supplies a weak method, the agent iterates for up to twelve hours, and a fresh container scores the final submission on a split the agent did not observe [1]. A keep is not merely a local note. It changes the ancestry of later work: the kept candidate becomes a parent, while a rejected candidate is excluded from the branch.

This becomes an accountability problem when artifacts move between agents or organizations. A downstream consumer may receive a final method and an informal experiment log but still be unable to answer three basic questions: what exact candidate was proposed, what evidence resolved the decision, and which exact bytes later work inherited. General provenance standards can represent entities, activities, agents, and derivation [2]; networked-agent architectures add discovery, authentication, and capability metadata [3]. Neither by itself defines the task-specific checks needed to re-evaluate a keep decision from exported files.

We present a portable record and offline verifier for that decision boundary. The contribution is deliberately narrower than manipulation-resistant reputation or malicious-agent con-

tainment. The evaluation is single-agent; it is not itself a decentralized or multi-agent network experiment. Its Track 3 relevance is that a downstream party can require a locally verifiable event before admitting a claimed improvement into its own lineage or reputation policy.

The paper makes three contributions:

- a decision-record contract that binds candidate and parent artifacts, evidence roles, measurements, resolution, and submitted lineage;
- an offline verifier that reports integrity, coverage, and protocol outcomes separately rather than collapsing missing evidence into success; and
- a source-pinned evaluation that reports a negative gate result and preserves the evidence class and precision of every primary row.

II. DECISION-PROVENANCE DESIGN

A. Auditable unit and trust boundary

The auditable unit is one keep, revert, provisional, or confirmation event over an exact candidate–parent pair. Each resolved decision records the statistic, direction, estimate, interval and coverage, deterministic method identifier, sample size, minimum effect, evidence paths and digests, and the rule state needed to reproduce the disposition. Statistical verdict and operational disposition remain separate: an inconclusive measurement may lead to a provisional state rather than silently becoming a keep.

The record is tamper-evident relative to supplied files. A digest identifies bytes; it does not establish truth, quality, authorship, independent witnessing, or that an event happened inside an untrusted container. The verifier therefore claims internal consistency and declared-protocol conformance, not runtime attestation.

B. Producer and verifier

The producer maps an exported job directory into a versioned record. It binds every cache-free method tree, parent link, kept or reverted state, submitted version, reward artifact, and decision evidence. The verifier recomputes file digests, lineage, submitted-version identity, intervals, and rule conformance without a hosted service. It fails closed on a changed artifact, a missing snapshot, an ambiguous submission, or an interval that cannot be recomputed from the bound results.

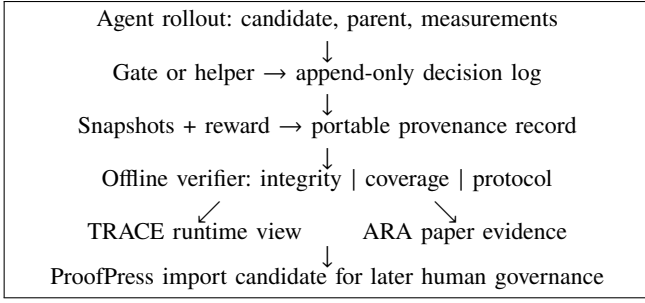


Fig. 1. Layers exchange data but do not inherit trust. TRACE expresses runtime decisions; the verifier checks bound files; ProofPress import is not approval; ARA packages the research argument.

Three outcome channels prevent a valid hash from being mistaken for a complete experiment. *Integrity* asks whether the supplied bytes match the record. *Coverage* asks whether required versions and decisions are present. *Protocol* asks whether the declared decision rules were followed. A record may pass one channel and fail another.

C. Interchange and governance layers

The decision log can be projected into a TRACE 0.5.1 session: the agent is the proposer, the gate is the resolver, and a confirmation event revises a provisional event. TRACE types the generic measurement while preserving profile-specific rule state as an identified extension. A downstream ProofPress adapter can import selected TRACE fields as external evidence, but import creates neither a claim nor an approval. These boundaries matter because provenance capture, source verification, decision representation, and authorization for later reuse are different operations.

The companion Agent-Native Research Artifact (ARA) stores the manuscript, falsifiable claims, experiment definitions, evidence pointers, and negative-result pivot as linked layers [4]. It is a paper-facing research package, not the runtime log or governance authority.

III. EVALUATION

A. Study design

We evaluate the system on `game2048_policy_search` using reduced windows and modified program overlays. These are not official RSI-Exam results. Within each block, one instrument-arm rollout and one helper-arm rollout use the same model-stage configuration. The instrument arm routes keep proposals through the statistical gate; the helper arm supplies provenance bookkeeping without that policy. The campaign manifest fixes stage configuration, randomized run order, primary endpoint, and missing-reward rules before the stages run.

The primary endpoint is instrument minus helper final sealed reward within a block. Secondary endpoints include record outcomes, recovered candidate–parent pairs, dispositions, and stop conditions. The ten blocks span four Haiku, three Sonnet, and three Opus pairs. The study was not sized

TABLE I
PRIMARY SEALED-REWARD DIRECTION BY MODEL STAGE

Stage	Blocks	Fav. I	Fav. H	Mean I–H
Haiku	4	2	2	−0.0186
Sonnet	3	1	2	−0.0171
Opus*	3	0	3	−0.1441
Total	10	3	7	–

*Block 1 bound; blocks 2–3 rounded summaries.

to estimate a population effect, so we report block directions and descriptive means, not an efficacy rate or significance test.

B. Source-bound reconciliation

A generator pins every paper input to repository revision 727b9b821d78 and refuses a working-tree source whose bytes differ. Haiku and Sonnet rows are parsed from machine-generated endpoint tables. For Opus block 1, each committed capsule records the source-precision hidden reward plus the SHA-256 of its `verifier/reward.json`. The repository lacks the generated Opus endpoint table and block 2–3 capsules; those four rewards are parsed from a committed pre-probe summary written before the subsequent probe and rounded to three decimals. The generated output preserves evidence class per row and refuses to reconstruct absent secondary results.

Evidence precedence is explicit: raw outputs and verifier receipts; generated per-trial tables; preregistered manifests and endpoint tables; committed pre-probe summaries; narrative handoffs; working notes. Lower-precedence prose cannot silently override a source table.

C. Results

Table I reports all ten primary blocks by stage. Three favor the instrument and seven favor the helper. Mean differences are negative in every stage; the Opus mean is approximate because two blocks use rounded inputs. This refutes the tested improvement claim under these conditions. It does not establish a general negative effect.

The three per-stage record tables enumerate 20 started trials and 18 verified records: 9 of 10 in the instrument arm and 9 of 10 in the helper arm. The two failures are visible `submitted_not_snapshotted` outcomes, one per arm. This matters independently of reward: the system did not silently reinterpret incomplete lineage as a valid record.

The provenance workflow also exposed an analysis-level failure. A ten-block narrative total existed before a canonical all-stage result file, while the Opus endpoint table and block 2–3 receipts were absent from the repository. The paper generator therefore carries per-row evidence class and a release-blocker list. Auditability here includes the ability to say which result is usable only for direction and which claim must remain unresolved.

IV. DISCUSSION

A. Negative efficacy, positive auditability

The gate is an evaluated policy, not the paper’s success criterion. Its loss is useful evidence for the provenance design: the same machinery that could have supported a favorable result instead preserves an unfavorable one, exposes incomplete records, and prevents a handoff summary from outranking its sources. The strongest claim is therefore about the record/verifier contract and evidence-precedence workflow.

Existing observations are compatible with a mechanism in which confirmation planning limits alter later search, but the comparison does not isolate that mechanism. We retain it as a hypothesis. A causal test would hold candidate evidence and compute fixed while varying the cap and remaining-window policy prospectively.

B. Implication for decentralized agent systems

NANDA architectures emphasize discovery, authentication, capability attestation, and interoperable agent ecosystems [3]. Our result supplies a narrower artifact-level primitive: before accepting an improvement, a consumer can require a record whose claimed lineage, evidence, and resolution are locally checkable. A network may then attach policy or reputation to explicit verifier outcomes rather than prose assertions. The record does not make reputation manipulation-resistant by itself, authenticate a producer, or prevent collusion and Sybil behavior.

C. Limitations

The study uses one task, small stages, modified programs, and reduced windows. Author-run evidence is not independent evidence. Digests detect changes relative to supplied files but do not prove correctness or runtime occurrence. Opus block 1

is reward-receipt-bound, but blocks 2 and 3 remain rounded summaries; Opus spend and complete secondary analyses are absent from the merged tree. ProofPress compatibility demonstrates a schema handoff, not adoption, approval, or governed reuse. Finally, the public workshop page does not state whether review is anonymous or whether author-controlled artifact links are allowed, so this review source remains anonymous and contains no public artifact link.

V. CONCLUSION

We introduced an offline-verifiable record for the keep-or-revert decisions that create lineage in self-improving-agent rollouts. The system binds artifacts and evidence, separates integrity, coverage, and protocol outcomes, and makes incomplete provenance visible. In ten reduced-window, modified-program comparisons, the tested gate does not improve the primary endpoint: three blocks favor the instrument and seven favor the helper. Reporting that negative result with its source-strength boundary is part of the contribution. For networked agents, the record offers a checkable event on which downstream governance and reputation mechanisms can operate without treating an agent’s narrative as proof.

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